Measurement of Internal Resistance in Lithium Batteries

Practical Implementation

Leonardo Chieco

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Measuring the internal resistance of an electrochemical cell is a fundamental aspect for evaluating its performance and state of health. This parameter directly affects the efficiency, the ability to deliver current, and the battery's behavior under load. However, obtaining a reliable measurement is not trivial, especially when trying to avoid the typical errors associated with direct-current techniques. This tutorial presents an approach based on alternating current excitation, which allows the internal resistance to be determined more accurately and, in a manner, consistent with the principles of electrochemical impedance spectroscopy (EIS). Starting from the basic theoretical concepts, a practical circuit solution is then developed, aiming to create a measurement system that is simple, yet effective and educationally meaningful.

***Warning**: working with lithium batteries can be potentially dangerous, especially when high currents or uncontrolled operating conditions are involved. Wiring mistakes, short circuits, or improper stress can lead to overheating, permanent damage, or hazardous situations. For this reason, it is strongly recommended to first study and validate the circuit through simulation, allowing you to analyze the system behavior and make any necessary modifications in complete safety before proceeding with a practical implementation.*

Enjoy the reading!

The Author
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1. Introduction

When working with lithium cells, one of the most significant parameters is the internal resistance, which serves as a direct indicator of the battery's State of Health (SOH) and, indirectly, of its remaining useful life. As the battery ages, internal electrochemical processes undergo changes: passivating layers form on the electrodes, charge-transfer resistance increases, and ion mobility in the electrolyte decreases. All of these effects lead to an increase in internal resistance.

An "aged" cell therefore exhibits larger voltage drops under load, higher heating, and a reduced ability to deliver power. For this reason, measuring internal resistance is one of the simplest and most effective tools for estimating the SOH and assessing how much a battery can still be used.

Over time, several methods have been developed to measure internal resistance, each with specific advantages and limitations. The most intuitive method is the direct current (DC) technique, which is based on applying a current step and measuring the resulting voltage change. In this case, Ohm's law is applied in the form $\Delta V/\Delta I$, observing the immediate response of the battery.

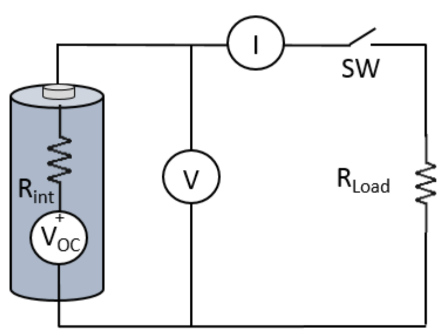

Figure 1: Measurement of Internal Resistance Using the Current-Step Method

In open-circuit conditions, no current flows through the circuit, so $V=V_{OC}$. When the switch SW is closed, the load R_{LOAD} draws a current I_{LOAD}, and $V=V_{LOAD}$.

The current I_{LOAD} also flows through R_{INT}, so by applying Kirchhoff's voltage law:

$$V_{OC} = R_{int}\, I_{LOAD} + R_{load}\, I_{LOAD} = R_{int}\, I_{LOAD} + V_{LOAD}$$

$$R_{int} = \frac{V_{OC} - V_{LOAD}}{I_{LOAD}}$$

In reality, the behavior of the cell is slightly more complex, as shown in the following figure.

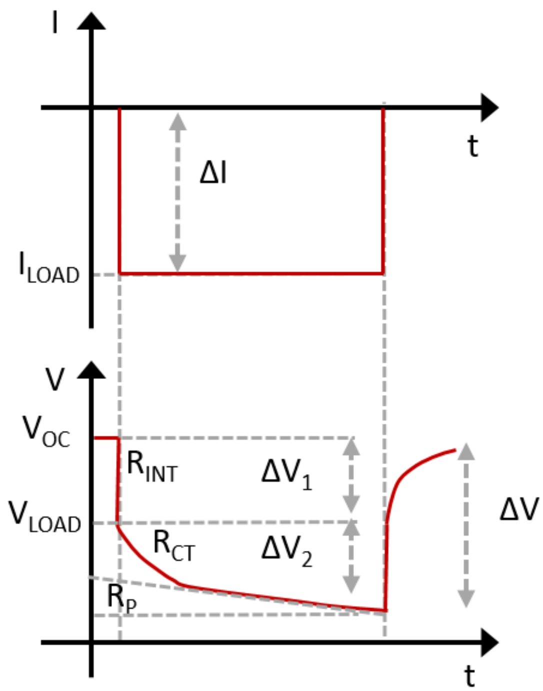

Figure 2: Current step response

When a current step is applied, the voltage does not decrease uniformly, but instead exhibits a characteristic behavior. At the instant of application, an immediate voltage drop is observed, followed by a slower and more gradual variation until a new quasi-steady-state value is reached. This shape is not random: it reflects the different physical phenomena occurring inside the battery over different time scales.

To interpret this response, an equivalent electrical model of the battery is often used, representing the electrochemical system through resistances and capacitances. Within this framework, three key parameters emerge: R_{INT}, Rct, and Rp.

R_{INT} is the so-called pure ohmic resistance. It is the most immediate contribution and represents all the “physical” resistances to current flow: the resistance of the electrodes, the electrolyte, the current collectors, and the contacts. It is responsible for the instantaneous voltage drop at the exact moment the current step is applied. From a mathematical point of view, it is the easiest parameter to estimate, as it can be obtained from the ratio between the immediate voltage change and the current step: $\Delta V/\Delta I$. It does not involve dynamic phenomena, and therefore it does not depend on time, but only on the physical state of the cell (temperature, state of charge, degradation).

Immediately after this initial drop, Rct, the charge transfer resistance, comes into play. This parameter is related to the kinetics of the electrochemical reactions occurring at the electrode–electrolyte interface. In other words, it describes how “easy” or “difficult” it is for electrons and ions to participate in the electrochemical reaction. Unlike R_{INT}, Rct does not appear instantaneously, but through a slower variation of the voltage, typically with an exponential trend. This is because it is associated with an RC-type behavior, i.e., a resistance in parallel with a capacitance (the so-called double-layer capacitance). Rct is particularly important because it is strongly sensitive to the battery’s state of health (SOH) and tends to increase with aging.

Finally, a parameter Rp, called polarization resistance, is often introduced. This term is more “global” and includes slower effects related to mass transport phenomena, such as ion diffusion within the electrodes and the electrolyte. If the voltage is observed over longer time scales (seconds or even minutes), it continues to vary slowly: this behavior is precisely attributed to polarization.

Measuring the internal resistance using this method requires instrumentation capable of generating well-defined, fast, and repeatable current steps, as well as acquisition systems with sufficient bandwidth and resolution to accurately capture both the instantaneous change and the subsequent evolution of the voltage. In particular, it is essential to clearly distinguish the change in slope of the response, which encapsulates the different resistive and dynamic components of the cell.

To overcome these limitations, more sophisticated methods are used, such as those based on time-varying signals. Among these, Electrochemical Impedance Spectroscopy (EIS) represents the most comprehensive, but also the most complex, approach.

Electrochemical Impedance Spectroscopy (EIS) is one of the most powerful tools for understanding what happens inside a battery, or more generally, any electrochemical system. The underlying idea is simple: instead of observing only the battery's response to direct current or voltage, as in classical charge-discharge tests, a small sinusoidal perturbation is applied, and the system's response is analyzed at different frequencies. This shift in perspective is crucial because it allows the internal physical phenomena to be separated, which is impossible with a static measurement.

Imagine applying a very small sinusoidal current to a battery and measuring the resulting voltage. If the battery were a simple resistor, voltage and current would be perfectly in phase. In reality, however, the voltage is phase-shifted relative to the current, and this phase shift carries valuable information. This is the basis of the complex impedance concept, which describes both the resistive and reactive components of the system. By varying the signal frequency, it is possible to "probe" the system under different operating conditions: at high frequencies, purely ohmic effects dominate, while at lower frequencies, slower phenomena such as electrochemical reactions and ion diffusion become apparent.

The most intuitive way to represent the results is with a Nyquist plot, where each point corresponds to a different frequency. The typical shape obtained for batteries is a semicircle followed by an inclined tail.

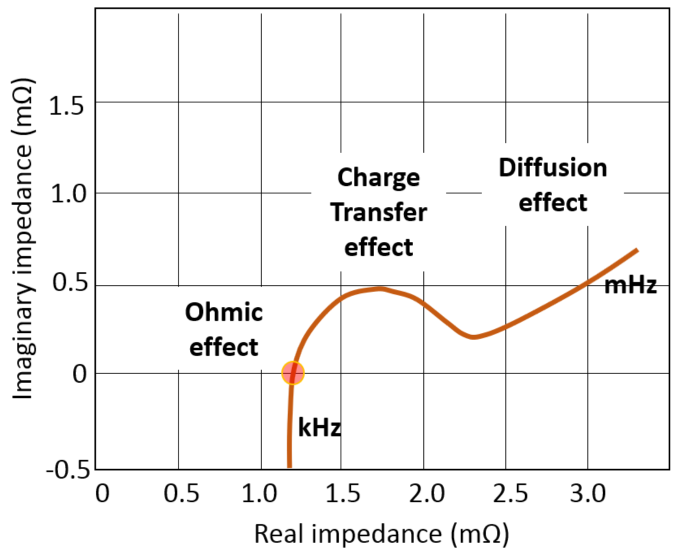

Figure 3: Example diagram of electrochemical impedance spectroscopy

This is not a simple graph, but a true “fingerprint” of the electrochemical system (cell): the intercept, where the imaginary part is zero, represents the pure internal resistance. The semicircle is associated with charge-transfer phenomena at the electrode–electrolyte interface, while the final tail (on the right) corresponds to ion diffusion within the active material. In other words, a single plot summarizes completely different physical processes.

Within this general framework, the AC internal resistance measurement at 1 kHz can be seen as a simplified, particular case of EIS. Instead of performing a full frequency sweep, a single point, typically at high frequency, is chosen, and the impedance is measured under that condition. The choice of 1 kHz is not arbitrary: at such “high” frequencies, slower phenomena, such as diffusion and electrochemical reactions, cannot follow the signal variation and are effectively “filtered out.” What remains is primarily the pure resistive component.

This approach has several practical advantages. The measurement is fast, does not require long acquisition times, and can be implemented with relatively simple circuits. Furthermore, since it is based on a small sinusoidal signal, it does not stress the battery and can even be performed during normal operation. However, it is important to be aware that this is a simplification: the measured value does not represent the battery’s complete behavior, but only a part of it. For advanced applications, such as accurate modeling or detailed diagnostics, a full EIS analysis is still required.

In the following section, a didactic circuit simulated in LTspice will be presented. The objective will be to implement a 1 kHz AC measurement on an equivalent lithium cell model, showing step by step how to extract the internal resistance from voltage and current. This will allow a direct connection between theory and practice, providing a solid foundation for more advanced developments.

2. Measurement of Internal Resistance

The block diagram of the system that we will simulate with LTSpice is shown in the following figure.

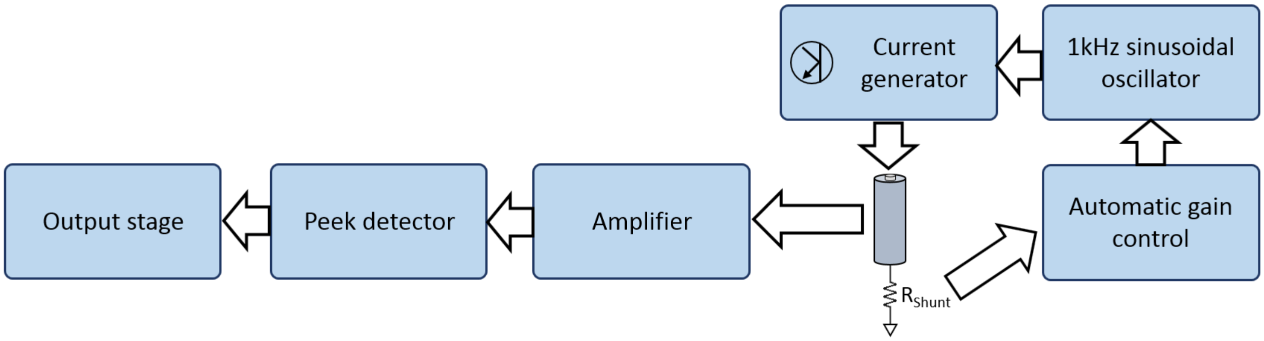

Figure 4: Block diagram of the proposed system

Several functional blocks are visible:

1. 1 kHz sinusoidal generator
2. Automatic amplitude control of the sinusoid
3. Current generator (active load)
4. Signal amplifier
5. Peak detector
6. Output stage

In the following sections, we will analyze the operation of each stage in detail.

2.2 Sinusoidal Wave Generator

Oscillators are devices capable of autonomously generating a sinusoidal output signal without the need for an external input. The only energy supplied to the system comes from the power source. In other words, an oscillator is a system that, although initially perturbed only by noise or initial conditions, can develop and maintain a free, undamped response over time.

This behavior is far from trivial. In real physical systems, dissipative phenomena are always present: resistances, dielectric losses, equivalent friction. For example, in a classical RLC circuit, after an initial perturbation, the oscillation gradually damps until it disappears. The initial energy is dissipated, and the system returns to equilibrium.

An oscillator, however, avoids this damping thanks to positive feedback. The fundamental idea is that the circuit does not simply allow the oscillation to evolve; it continuously regenerates it, compensating exactly for internal losses. In this way, the dissipated energy is restored at each cycle, and the oscillation amplitude remains constant.

To understand the mechanism, it is useful to describe the system in terms of functional "black box" blocks. Consider an amplifier with transfer function A, followed by a feedback network B, which feeds part of the output signal back to the input.

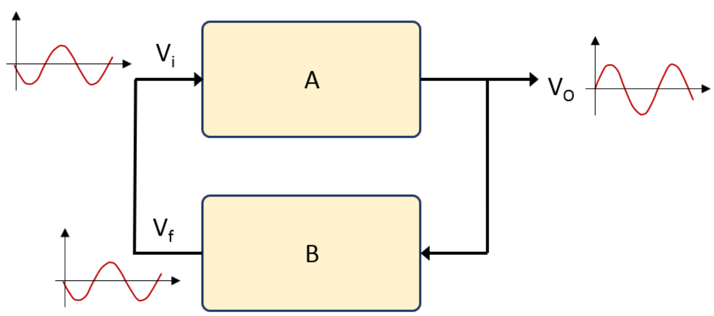

Figure 5: Black-box diagram of an oscillator

Suppose that, due to noise or a small perturbation, a sinusoidal signal V_i appears at the input of block A. Passing through block A, this signal is amplified and phase-shifted: the output V_O remains sinusoidal at the same frequency (if the system is linear), but with modified amplitude and phase.

Next, the signal V_O passes through the feedback network B, which introduces a further amplitude variation and phase shift. The resulting signal, V_f, is fed back to the input.

The crucial condition is as follows: if the system is designed such that the returned signal V_f is identical to the initial signal V_i, the circuit enters a self-sustaining condition. The signal regenerates each cycle without attenuating or growing indefinitely.

This condition can be compactly expressed in the frequency domain. Denoting the angular frequency of the sinusoidal signal as ω, the requirement is:

$$A(\mathrm{j}\omega)\, B(\mathrm{j}\omega) = 1$$

This is the compact form of the Barkhausen criterion, which establishes the conditions for the initiation and maintenance of oscillations.

Separating magnitude and phase, the criterion translates into two fundamental conditions:

- The overall phase of the loop must be zero (or a multiple of 2π): $\angle A(\mathrm{j}\omega)\, B(\mathrm{j}\omega) = 1$
- The loop gain magnitude must be unity: $|A(\mathrm{j}\omega)\, B(\mathrm{j}\omega)| = 1$

The first condition ensures that the feedback signal is in phase with the input, thus adding constructively. The second condition guarantees that the amplitude remains constant: neither attenuated (oscillation dies) nor amplified indefinitely (oscillation diverges to saturation).

A phase-shift oscillator using an operational amplifier is one of the most elegant ways to generate a sinusoidal signal without using complex active components.

In this case, block A introduces a 180° phase shift (inverting amplifier), while network B, implemented with RC cells, provides the other 180°, thus satisfying the Barkhausen phase condition. The 180° phase-shift network is composed of three cascaded RC cells. A single RC network cannot provide a 180° phase shift without completely attenuating the signal, so three identical stages are used, each contributing about 60° of phase shift at the oscillation frequency. When the frequency is correct, the sum of the three RC phase shifts reaches 180°, which, added to the 180° of the inverting op-amp, yields the required 360°.

As already noted, the signal returning to the input must have sufficient amplitude to compensate for losses in the RC network. In a classical three-stage RC oscillator, the attenuation is such that the amplifier must have a minimum gain of approximately 29 for the oscillation to start and sustain itself.

The oscillation frequency is not arbitrary but depends on the R and C values of the network. In the symmetric case (all resistances equal and all capacitances equal), the frequency is:

$$f = \frac{1}{2\pi RC\sqrt{6}}$$

This relation shows that the frequency is determined by the RC time constant and a factor related to the three-stage structure. It is not merely a filter but a network designed to achieve the desired phase shift at the oscillation frequency.

The following figure shows a classical phase-shift oscillator schematic.

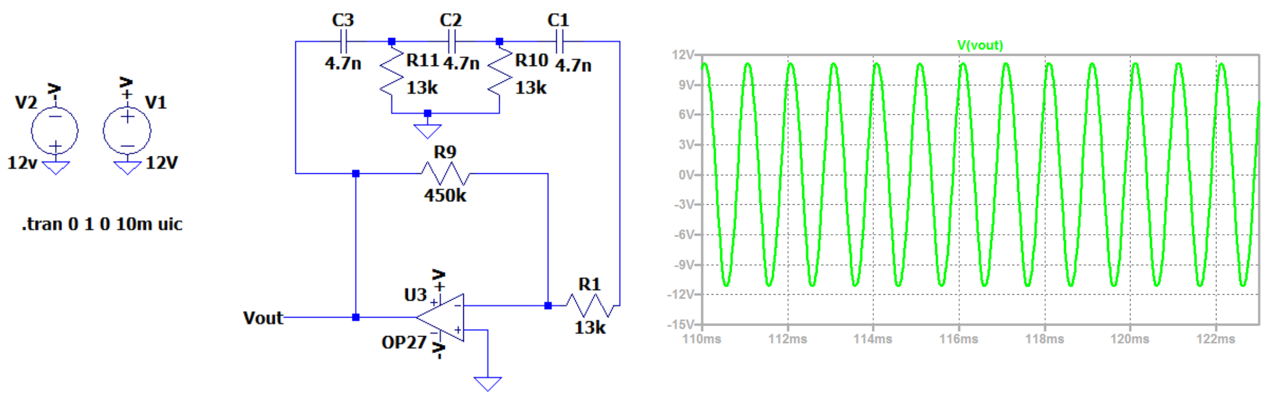

Figure 6: Schematic and simulation of a phase-shift oscillator

With the components used, we have:

$$f = \frac{1}{2\,\pi RC\sqrt{6}} = \frac{1}{6.28 \cdot 13 \cdot 10^{3} \cdot 4.7 \cdot 10^{-9} \cdot 2.45} \cong 1kHz$$

For reasons that will become clearer later, it is necessary to have a sinusoid that does not oscillate around zero but remains always positive, and with adjustable amplitude. To achieve this, the basic oscillator circuit was modified by introducing two key elements: a voltage offset and an amplification stage. Specifically, op-amp U10 generates a reference voltage approximately equal to half the supply, acting as a virtual ground. This voltage is applied to both the phase-shift network and the non-inverting input of U3. In this way, the generated sinusoid oscillates around this reference level rather than zero volts, remaining always positive. Stage U1 controls the signal amplitude, adapting it to the downstream system requirements.

The resulting circuit, shown in the figure below, generates a sinusoid at ~1 kHz with a swing from ~0 V to 11 V at U3 output, while at U1 output a scaled version is obtained, with amplitude roughly between 0 V and 1.2 V.

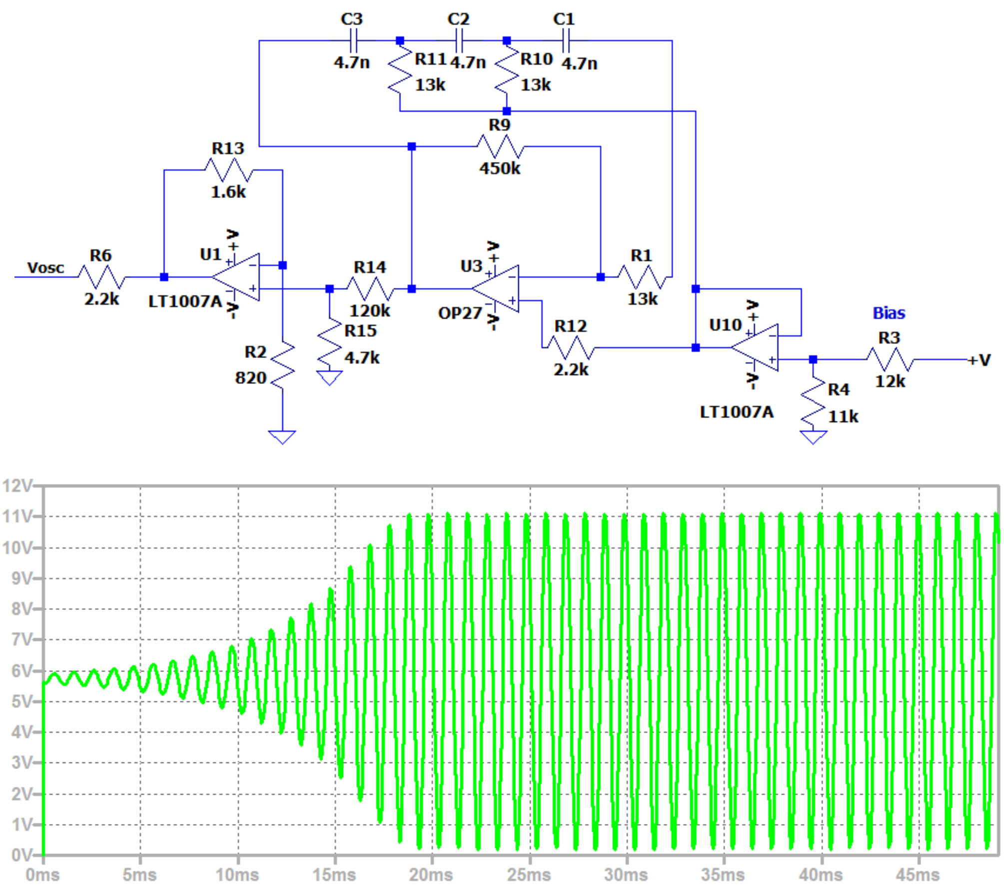

Figure 7: Schematic and simulation of the oscillator stage

2.3 Voltage-to-Current Converter (Active Load)

The sinusoidal voltage from the oscillator stage is converted into a current to be drawn by the electrochemical cell under test. For this reason, the sinusoid must remain always positive.

The circuit used is a classic voltage-to-current converter implemented with an operational amplifier and an NPN BJT transistor. The idea is to exploit the op-amp's ability to force its output to maintain equal voltages at its inputs, using the transistor as a power element to supply the required current.

To understand the operation, imagine that the sinusoidal voltage V_{OSC} is applied to the non-inverting input of the op-amp. The inverting input is connected to a sensing resistor R_{SENSE}, through which the current flows toward the cell.

Since the op-amp operates in negative feedback, it enforces:

$$\mathrm{V}- \approx \mathrm{V}+= \ V_{OSC}$$

Consequently, the voltage across R_{SENSE} equals V_{OSC}, and the current through it, which coincides with the current supplied to the cell, is:

$$I_{OUT} = \ V_{OSC} \ R_{SENSE}$$

The output current depends only on the input voltage and the resistor value, and is therefore independent of the load, at least within the circuit's operational limits. Even if the cell impedance changes, the circuit continues to force the same current by automatically adjusting the output voltage.

The NPN transistor acts as the active power element, supplying the required current, while the op-amp handles control. The op-amp does not directly provide current to the cell but drives the transistor base to impose the desired current. This approach leverages the high gain and precision of the op-amp for control, leaving the transistor to handle power.

For higher currents (on the order of amperes), a Darlington configuration of two transistors is used, increasing the overall current gain, reducing the op-amp's effort, and improving stability and linearity.

The capacitor C5 smooths the op-amp's reaction to instantaneous changes, preventing oscillations or overshoot. It essentially acts as an integrator.

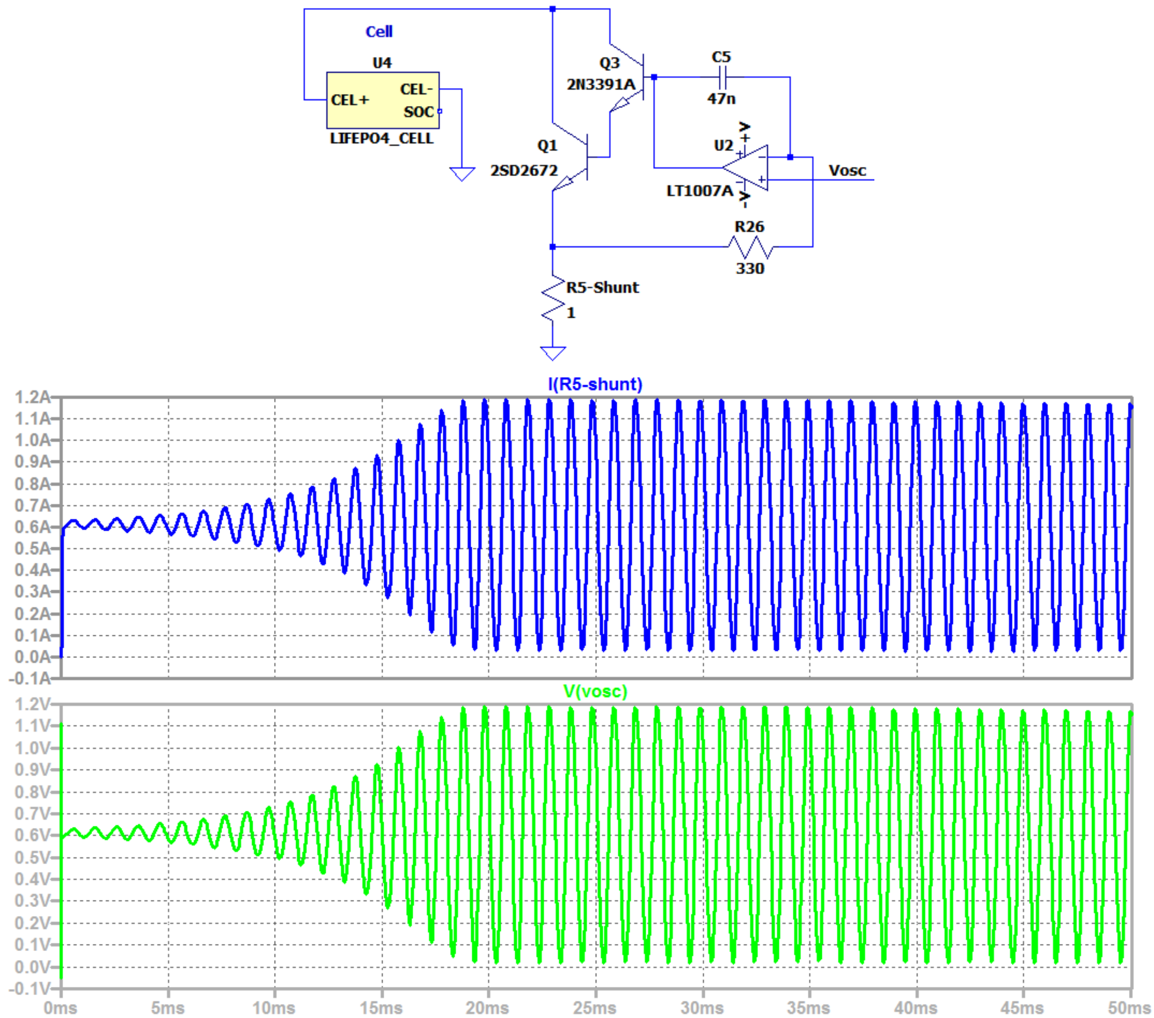

Figure 8: Schematic and simulation of the voltage-to-current converter stage

2.4 Automatic Gain Control

In this project, we aim to have the electrochemical cell draw a sinusoidal current at 1 kHz with a maximum amplitude of exactly 1 A. This requirement is far from trivial, as it implies that the circuit must be capable of generating a controlled, stable, and well-defined current even when the cell's impedance and the surrounding conditions vary.

The measurement of internal resistance (or, more generally, impedance) depends directly on the accuracy with which the applied current is known. If the current amplitude varies, the resulting measurement will also be affected by error. For this reason, it is essential to introduce a form of feedback that keeps the current amplitude constant, regardless of load variations.

The generated sinusoid is intentionally produced with a slightly higher amplitude than required, so that an amplitude control system can be applied to reduce it precisely to 1 A.

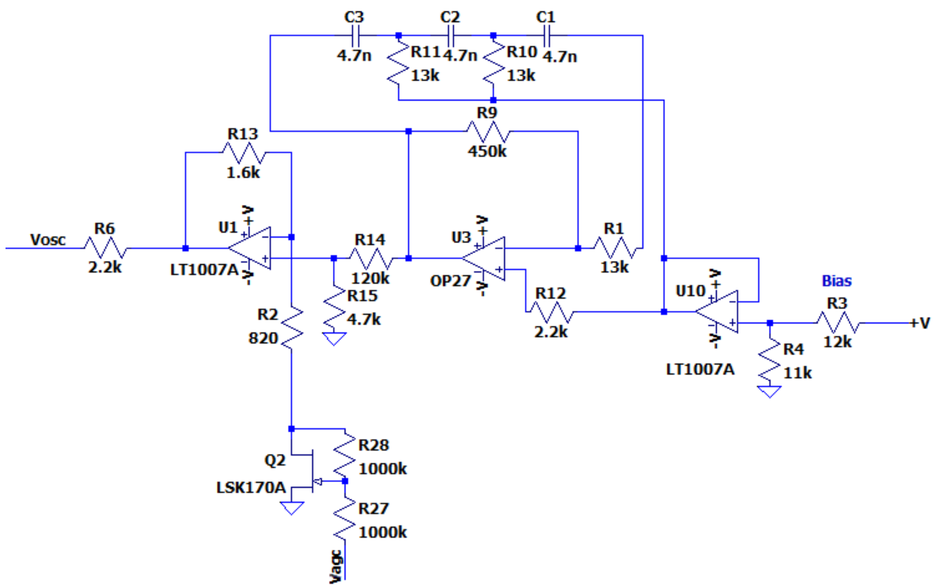

Figure 9: Schematic Diagram of the Automatic Gain Control Using a JFET

To achieve this result, it is necessary to introduce a mechanism that automatically regulates the signal amplitude, namely an AGC (Automatic Gain Control) stage applied to amplifier U1. The underlying idea is to transform the system into a closed-loop control, where the current amplitude is no longer determined solely by the passive components, but is continuously corrected based on the actual delivered value.

The principle used is particularly elegant: an N-channel JFET is employed as a voltage-controlled resistor. This device exhibits a resistance between drain and source that depends on the voltage applied between gate and source. When $V_{GS} = 0$, the JFET is in its maximum conduction state and behaves approximately like a short circuit, thus being practically transparent in the circuit. As the V_{GS} voltage becomes more negative, the channel narrows, and the resistance between drain and source increases progressively.

By inserting the JFET in series with the feedback resistor R2 of amplifier U1, a variable gain is obtained: when the JFET is "open" (low resistance), the gain is maximum; when the JFET "closes," it introduces an additional resistance that reduces the amplifier's overall gain. In this way, by acting on the gate voltage, it is possible to directly control the amplitude of the sinusoidal signal and, consequently, the current delivered to the cell.

The system's goal is to maintain the maximum current at 1 A. To achieve this, the current must be measured and compared with a reference value. The current flowing through the cell passes through a shunt resistor R_{SHUNT}, which converts it into a proportional voltage. For a current of 1 A, a voltage of 1 V develops across the shunt, which then becomes the value to be maintained constant.

At this point, the actual control system comes into play. The voltage across the shunt is sinusoidal, but what matters is its amplitude. To extract this information, a peak detector stage (U6) is used, which charges capacitor C6 to the maximum value of the voltage V_{SHUNT}. This produces a DC voltage proportional to the current amplitude.

This voltage is then compared with a fixed 1 V reference, representing the desired setpoint. The comparison is made using a unity-gain differential stage (U5), which generates an error signal: if the current is too high, the signal will be positive; if it is too low, the signal will be negative.

To make the control effective even in the presence of small errors, an inverting integrative controller is implemented using operational amplifier U7. The integrator accumulates the error over time, amplifying even very small deviations and bringing the system precisely to the desired value. This is particularly important because it allows elimination of steady-state error and maintains a stable current amplitude.

The output of the inverting integrator generates the control voltage V_{AGC}, which drives the gate of the JFET. Depending on this voltage, the JFET changes its resistance and thus the gain of amplifier U1, closing the regulation loop. Diodes D1 and D2 serve to limit the control signal, ensuring that the JFET gate is driven only with negative voltages, a condition necessary for the proper operation of the device.

The complete circuit is shown in the following figure.

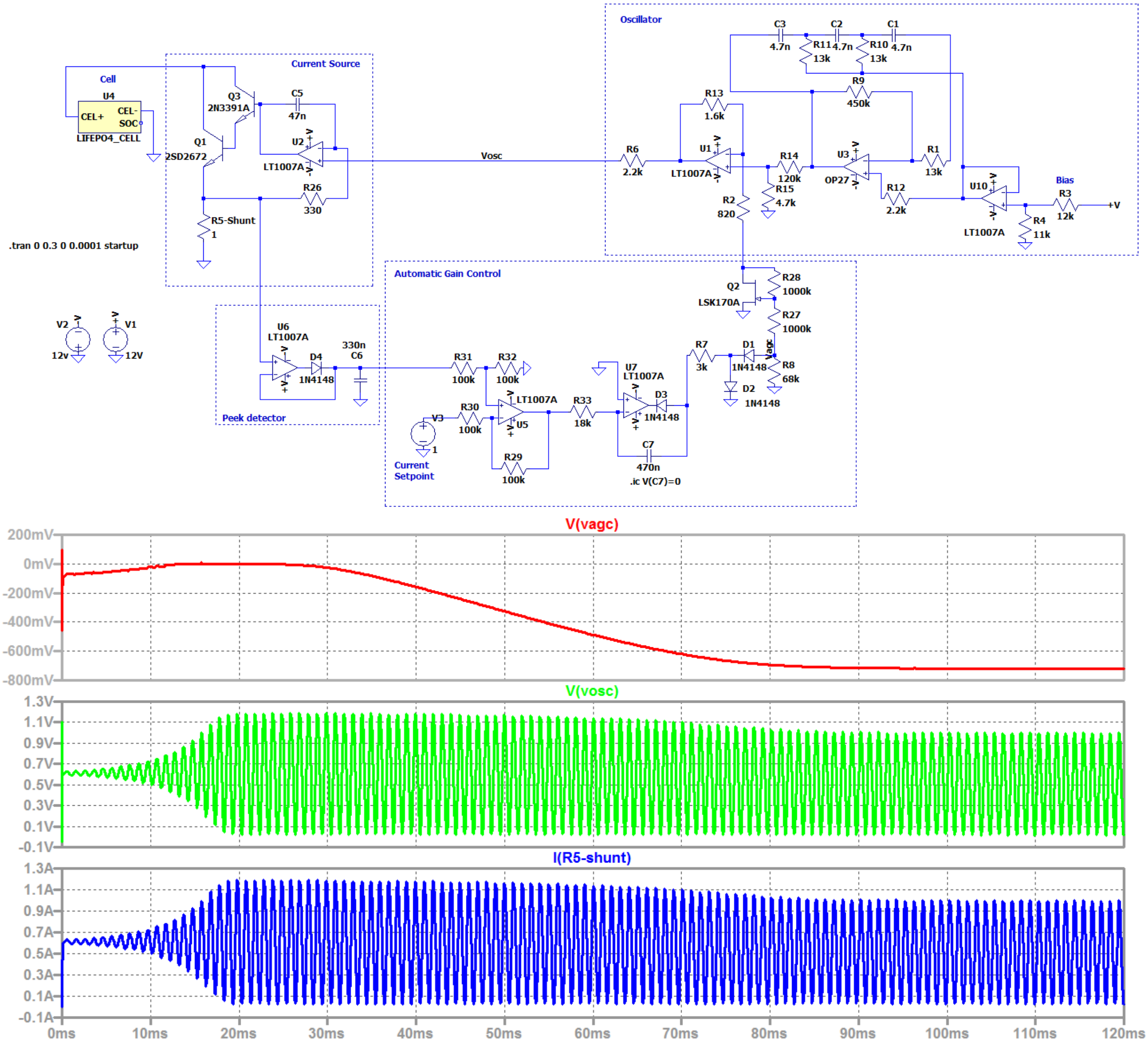

Figure 10: Complete Current Generation Stage with Amplitude Control

2.5 Signal Amplifier

Let us assume that the internal resistance of the electrochemical cell is on the order of 1 mΩ. If it is excited with a sinusoidal current with a peak-to-peak amplitude of 1 A, the voltage developed across its terminals will be extremely small, about 1 mV peak-to-peak. This is a very weak signal, which cannot be reliably measured directly, neither with a standard multimeter nor, even less, with an analog-to-digital converter without proper conditioning.

It is therefore necessary to introduce a precision amplification stage, capable of bringing the signal to levels compatible with measurement instruments, without degrading its informational content. This step is far from trivial, because when working with signals on the order of millivolts (or lower), errors introduced by the electronics can become comparable in magnitude to the signal itself.

For this reason, it is essential to use operational amplifiers characterized by very low offset, low noise, and minimal thermal drift. These three characteristics directly determine the quality of the measurement.

A very suitable example for this type of application is the INA849 operational amplifier, specifically designed for low-level precision measurements.

The low output offset means that even in the absence of an input signal, the operational amplifier does not introduce significant spurious voltage at the output. Ideally, with zero input, the output should be zero; in reality, due to internal imperfections, a small voltage called offset appears. For example, if an amplifier with a 1 mV offset is used to amplify a signal that is also 1 mV, a 100% error is introduced, making the measurement unusable. This is why, in these applications, the offset must be at least one or two orders of magnitude smaller than the signal to be measured.

Thermal drift represents the variation of the offset with temperature. Even if the initial offset is very low, a significant variation with temperature can compromise measurement stability over time. For instance, a drift of 5 µV/°C may seem negligible, but over a 20 °C change it results in a 100 µV error, which becomes relevant if the useful signal is on the order of millivolts. Zero-drift amplifiers are designed precisely to minimize this effect, keeping the offset practically constant with temperature changes.

The third aspect, often the most critical, is noise. Even in the absence of a signal, the amplifier’s output is not perfectly stable, but exhibits random fluctuations due to internal physical phenomena (thermal noise, flicker noise, etc.). This noise superimposes on the useful signal and limits the minimum measurable resolution.

Noise is typically specified as a spectral density, for example in nV/√Hz. This value indicates how much noise is present per unit bandwidth. To obtain the total noise, this density must be integrated over the band of interest. For example, if an operational amplifier has 10 nV/√Hz noise and a 1 kHz bandwidth is observed, the effective noise will be approximately:

$$V_{noise} = 10\,\frac{nV}{\sqrt{Hz}} \cdot \sqrt{1000\,Hz} \approx 316\,\mathrm{nV}$$

If the signal to be measured is on the order of 1 mV, this noise is negligible; but if it drops to tens of microvolts, it becomes a real limit to measurement precision.

From a practical standpoint, noise can be observed by connecting the amplifier input to ground and measuring output fluctuations with an oscilloscope or acquisition system. In this case, what is seen is not a deterministic signal, but a “cloud” of random variations around zero. Their statistical amplitude (e.g., RMS or peak-to-peak) represents the system noise.

The following figure shows the amplification stage.

Capacitor C8 decouples the DC component of the signal (open-circuit voltage of the cell). The total gain of this stage is:

$$G_{tot} = G_1 \cdot G_2 = \left(-\frac{R_{17}}{R_{16}}\right) \cdot \left(-\frac{R_{21}}{R_{20}}\right) = \left(-\frac{680\ k}{12\ k}\right) \cdot \left(-\frac{150\ k}{15\ k}\right) = 567$$

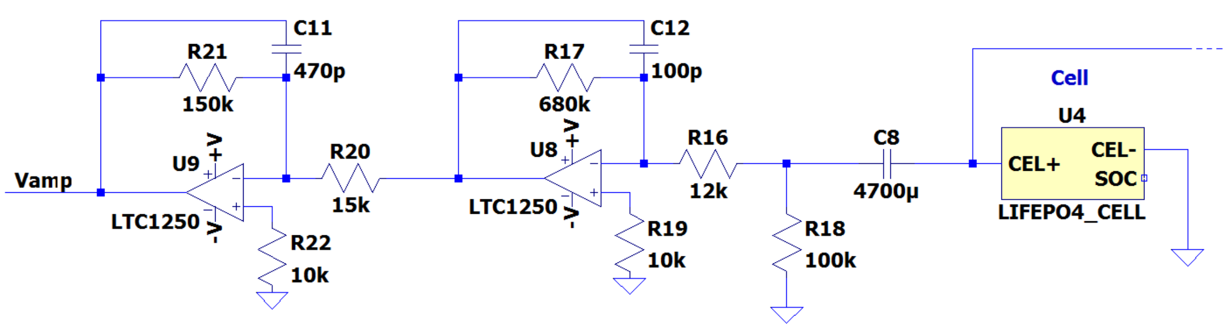

Figure 11: Amplification stage

Capacitors C11 and C12 reduce the bandwidth of U9 and U8; in fact, together with resistors R21 and R17, they form a low-pass filter with a cutoff frequency equal to:

$$f = \frac{1}{2\pi\,\mathrm{C}\,\mathrm{R}}$$

$$f_{U8} = \frac{1}{2\pi \cdot 100 \cdot 10^{-12} \cdot 680 \cdot 10^{3}} = 2341\ Hz$$

$$f_{U9} = \frac{1}{2\pi \cdot 470 \cdot 10^{-12} \cdot 150 \cdot 10^{3}} = 2258\ Hz$$

2.6 Peak Detector

The peak detector stage extracts the maximum of the amplified signal. Specifically, it detects the negative peaks.

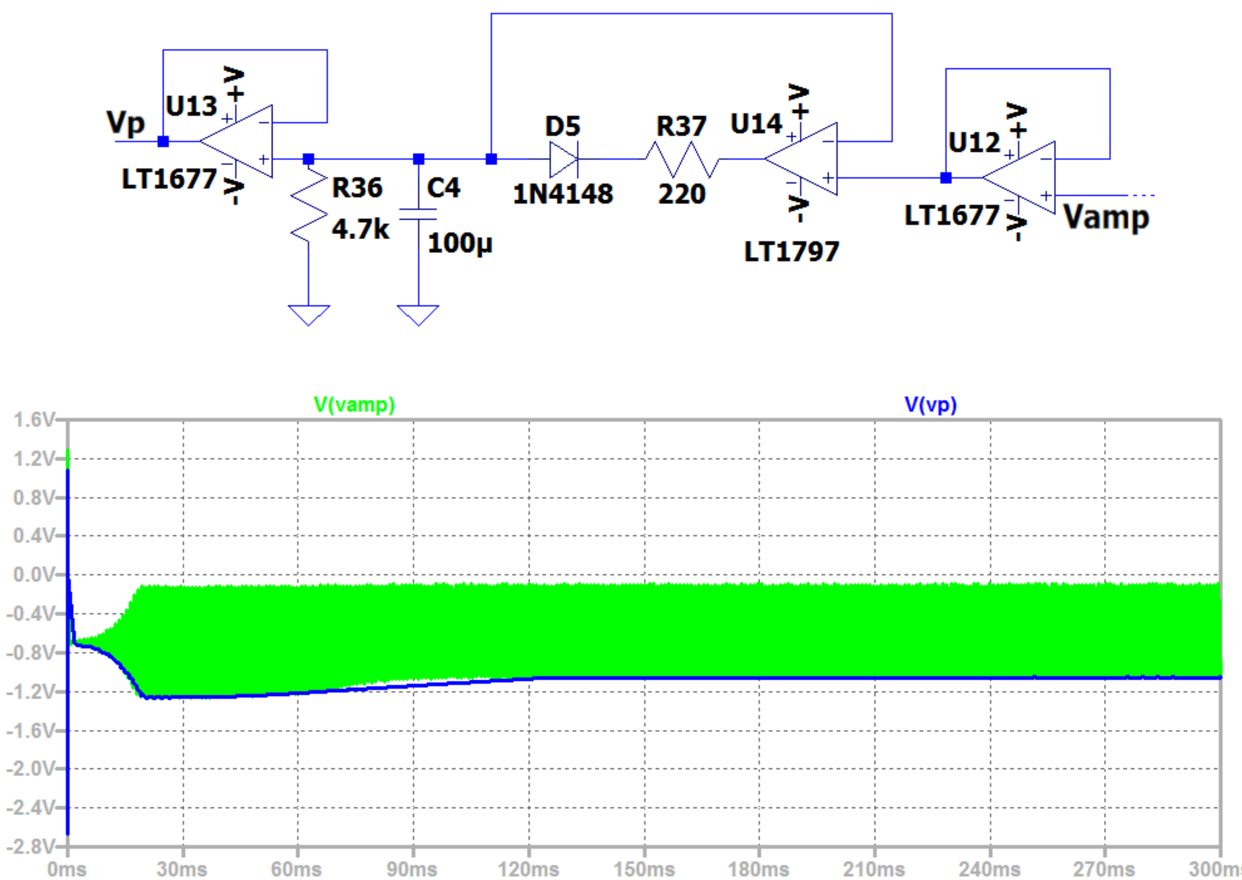

Figure 12: Schematic and simulation of the peak detector stage

2.7 Output Stage

The final stage of the system is responsible for performing a fine calibration of the overall gain, in order to obtain a direct and easily interpretable relationship between the measured quantity and the output signal. In particular, the circuit is adjusted so that the final response is 1 mΩ per volt, meaning that 1 V at the output corresponds to an internal resistance of 1 mΩ.

Capacitor C9 sets a cutoff frequency around 90 Hz, filtering out any residual AC components.

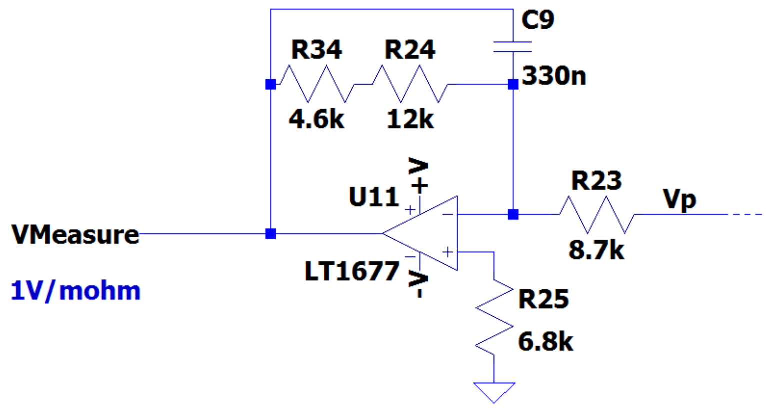

Figure 13: Output stage

2.8 Complete circuit

The following figure shows the complete schematic of the circuit with all interconnected stages.

In the appendix, you can find the electrochemical cell model to import into LTSpice.

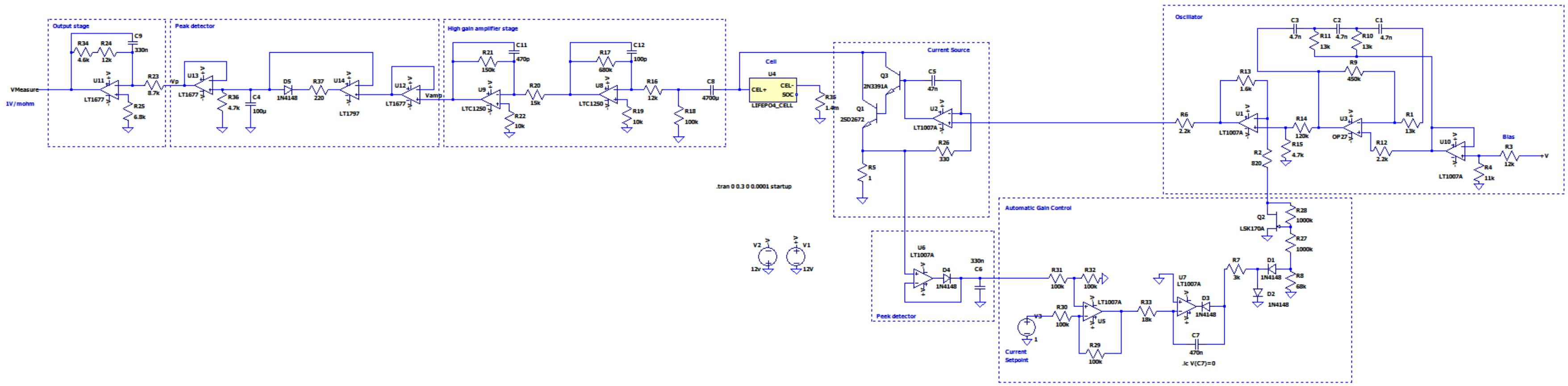

Figure 14: Complete circuit

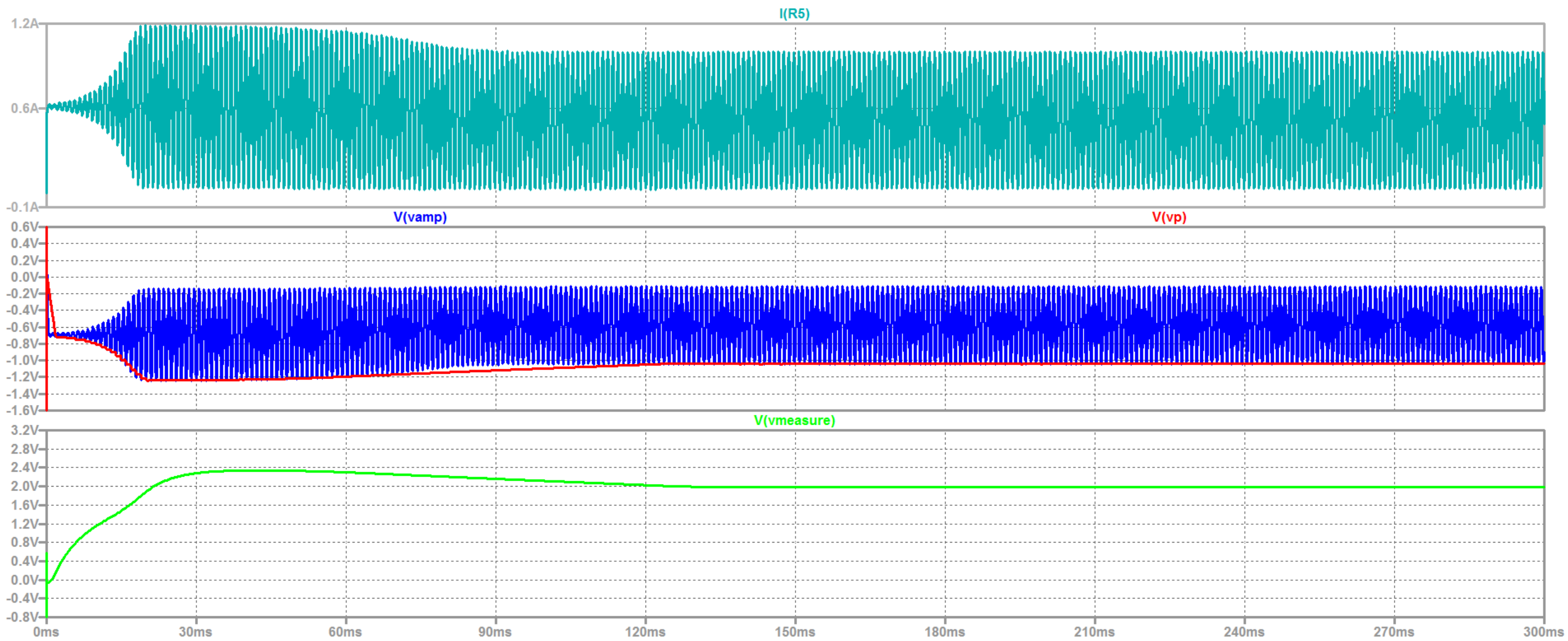

Figure 15: Main waveforms

3. Conclusions

In this tutorial, we have seen how it is possible to measure the internal resistance of an electrochemical cell using an alternating current approach, addressing the problem from both a theoretical and practical perspective. The use of a sinusoidal excitation at a known frequency allows for effective isolation of the purely resistive contribution of the cell, avoiding the typical ambiguities of DC measurements and approaching, albeit in a simplified form, the principles underlying electrochemical impedance spectroscopy.

Throughout the work, it was shown how, starting from a relatively simple architecture, comprising an oscillator, a voltage-to-current conversion stage, and a high-precision measurement chain, it is possible to realize a complete system capable of characterizing the cell with good accuracy. A key element of the project is the control of the injected current, achieved through feedback and automatic gain regulation, which allows the amplitude of the excitation to remain constant, thus ensuring stable, repeatable measurements that are independent of load variations.

It is important to emphasize that this is a project with an educational purpose, which inevitably leaves room for improvement and optimization. The main objective, however, was not to create a definitive instrument, but rather to build a clear pathway that starts from theoretical fundamentals and leads to a concrete circuit implementation, intentionally simple yet at the same time meaningful from a design perspective.

4. Appendix

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*==========================================================================================
* LIFEPO4_CELL subcircuit for LTspice
*
* Based on:
*
* SPICE Model Implementation for LiFePO4 Cell
*
* DOI:10.1109/ISFEE51261.2020.9756177
*
* Conference: 2020 International Symposium on Fundamentals of Electrical Engineering (ISFEE)
*==========================================================================================

.SUBCKT LIFEPO4_CELL CEL+ CEL- SOC
.PARAM NOMINAL_CAPACITY=180 INTERNAL_RESISTANCE=0.6m INITIAL_SOC=0.5

* ---------------------------
* Open circuit voltage vs SOC
* ---------------------------
E_OCV CELP CELN TABLE {V(SOC)} =(0,2.5) (0.03,2.8) (0.05,2.9) (0.075,3) (0.1,3.1) (0.12,3.17) (0.2,3.2)
(0.3,3.23) (0.4,3.28) (0.5,3.29) (0.6,3.3) (0.65,3.3) (0.7,3.31) (0.76,3.32) (0.8,3.33) (0.9,3.34)
(0.95,3.35) (0.97,3.37) (0.98,3.38) (1,3.5)

V_CSENSE CELN CEL-
R_INT CELP CEL+ {INTERNAL_RESISTANCE}
R_CONV CEL+ CEL- 1e12

CCELL_CAPACITY SOC 0 {3600*NOMINAL_CAPACITY}

G_BAT_CURRENT 0 SOC VALUE = { I(V_CSENSE) * IF(V(SOC)<0 & I(V_CSENSE)<0, 0, 1) * IF(V(SOC)>1 &
I(V_CSENSE)>0, 0, 1) }

RSOC SOC 0 1e12

.IC V(SOC)={INITIAL_SOC}

.ENDS
```